

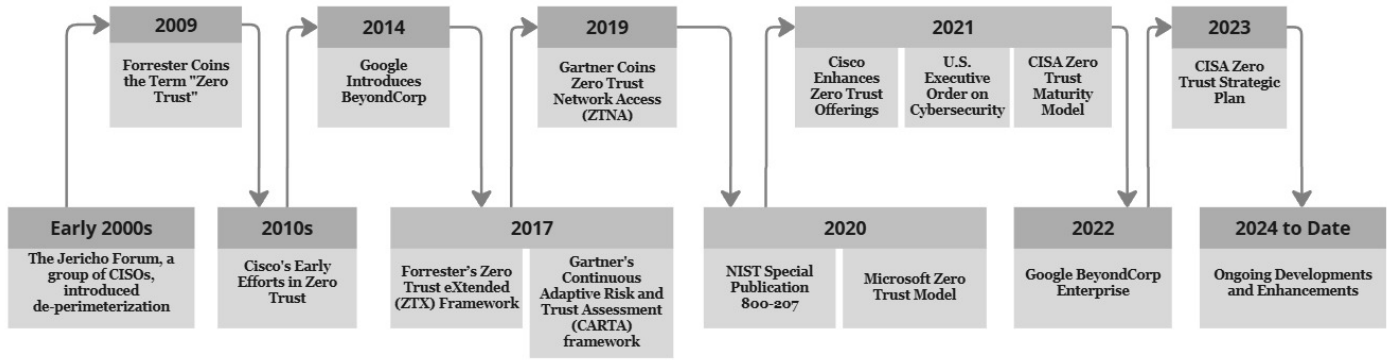


Figure 2: ZTA Historical Timeline

development of specific frameworks, such as CISA's Zero Trust Maturity Model [57]. Similarly, the release of special publication 800 - 207 on ZTA in August 2020 by National Institute of Standards and Technology (NIST) [58] may have facilitated academic research and practical implementations, leading to increased scholarly activity.

- **Escalation of Cyber Threats and High-Profile Cyber-attacks:** The period between 2020 and 2023 experienced a notable increase in cyber threats which include ransomware attacks, data breaches, e.t.c. High-profile incidents highlights the inadequacies of traditional security approaches. These include the SolarWinds cyberattack in 2020, the Colonial pipeline ransomware attack in 2021, the Microsoft exchange server exploit in 2021, the MOVEit file transfer vulnerabilities 2023, e.t.c. These threats prompted organizations to adopt more resilient security architectures like Zero Trust to better protect sensitive data and critical infrastructure, and may have influence the surge in publications within this time-frame.

4 Methodology

This section describes the approach used to carry out a Systematic Literature Review (SLR) with the goal of investigating Zero Trust architecture (ZTA) in light of its applications, issues and challenges, and enabling technologies. An SLR involves a methodical and structured approach to identifying, assessing, and interpreting relevant existing research with an aim to answer some research questions [59]. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting criteria is followed in this study. PRISMA framework is widely used in the literature and it provides a structured approach for conducting an SLR in a reproducible and transparent manner. Consequently, the methodologies employed to carry out the SLR are highlighted in this section. It comprises the questions that the study seeks to answer, the database from which the data is obtained, the search strategy, the procedure for screening and selecting the articles

based on inclusion and exclusion criteria, and the study selection and characteristics.

4.1 RESEARCH QUESTIONS AND OBJECTIVES

Developing research questions and objectives is essential to carrying out an SLR because it gives it a clear and deliberate focus that directs several processes, such as study selection and data analysis. The following research questions and objectives drive the process of conducting the SLR in this study:

Research Questions (RQs):

- RQ1: What are the applications of Zero Trust Architecture across various industries and sectors?
- RQ2: What are the issues and challenges faced in the adoption and implementation of Zero Trust Architecture?
- RQ3: What are the emerging technologies that contribute to Zero Trust Architecture?

Research Objectives (ROs):

- RO1: To identify and analyze various applications of Zero Trust Architecture across industries and sectors.
- RO2: To investigate the key issues and challenges encountered in the adoption and implementation of Zero Trust Architecture.
- RO3: To explore the emerging technologies that enable or enhance Zero Trust Architecture.

4.2 INFORMATION SOURCES/DATABASES

Although publications such as journal articles, conference papers, review/survey papers, etc can be found in a variety of databases, three popular databases — Scopus, Web of Science (WoS), and IEEE Xplore — were chosen as information sources for this study.

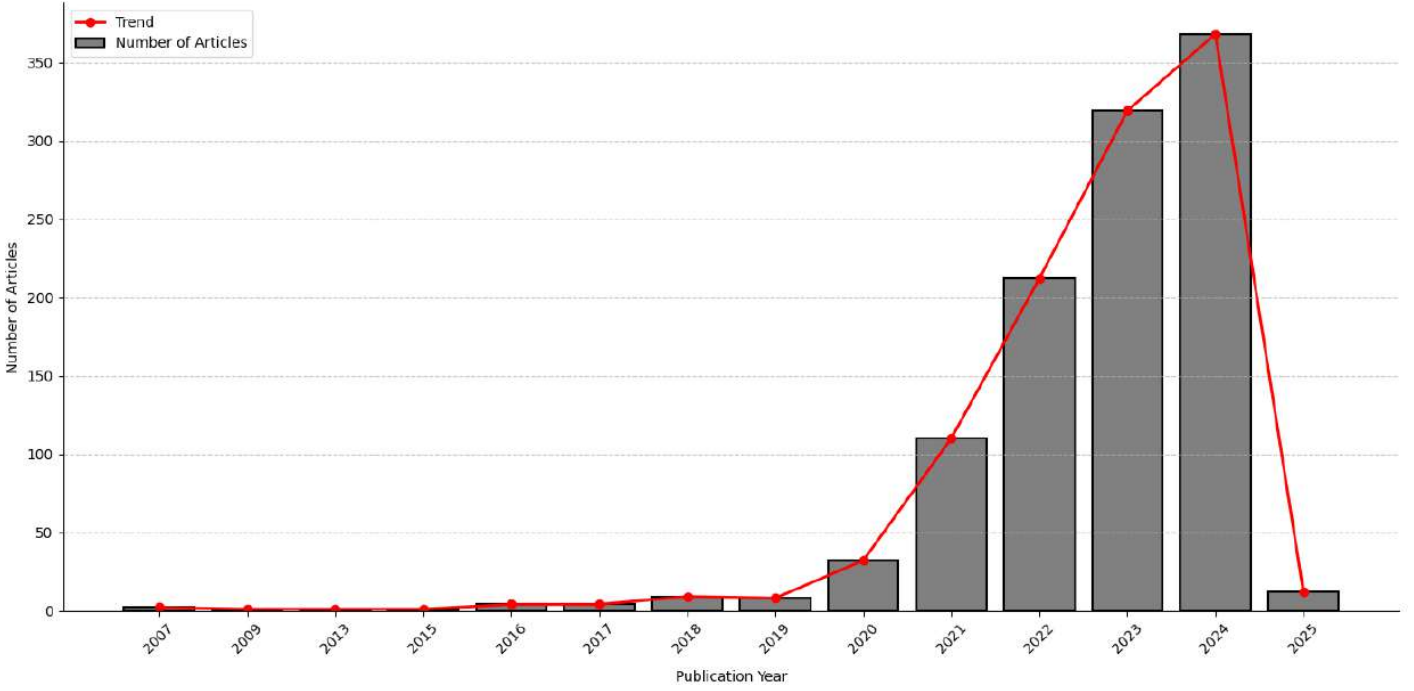


Figure 3: Publication Trend of Articles on ZTA over the Years (Scopus Database)

Their extensive article collection, general recognition and reputation in the academic community, and efforts to reduce the quantity of duplicate articles are the reasons behind this. The databases were thoroughly searched to find relevant articles for the study.

4.3 SEARCH STRATEGY/SEARCH TERMS

This step represents a crucial phase in conducting this study as it describes the strategy employed to identify the research articles reviewed based on the questions and objectives of the research. The selected databases differ slightly in the way they are queried. In essence, while publications can be searched in the Scopus database through their "Title-Abstract-Keyword" together, these are separate in WoS database and they have to be combined if needed. Therefore, in WoS database we repeated the search terms to search for the articles through the title and abstract with assumption to get the same effect and results as Scopus database. The following are the respective search term/string used in each of the databases to retrieve the relevant articles:

In Scopus database, we used:

TITLE-ABS-KEY ("Zero Trust Architecture" OR "Zero-trust Architecture" OR "Zero-trust" OR "Zero Trust" OR "Zero Trust Security" OR "Zero-trust Security") AND PUBYEAR > 2016 AND PUBYEAR < 2025 AND (LIMIT-TO (SUBJAREA, "COMP") OR LIMIT-TO (SUBJAREA, "ENGI")) AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "cp") OR LIMIT-TO (DOCTYPE, "cr") OR LIMIT-TO (DOCTYPE, "re"))

AND (LIMIT-TO (PUBSTAGE, "final")) AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (SRCTYPE, "p") OR LIMIT-TO (SRCTYPE, "j"))

While, in Web of Science (WoS), we used:

"Zero Trust Architecture" OR "Zero-trust Architecture" OR "Zero-trust" OR "Zero Trust" OR "Zero Trust Security" OR "Zero-trust Security" (Title) or "Zero Trust Architecture" OR "Zero-trust Architecture" OR "Zero-trust" OR "Zero Trust" OR "Zero Trust Security" OR "Zero-trust Security" (Abstract) and "Zero Trust Architecture" OR "Zero-trust Architecture" OR "Zero-trust" OR "Zero Trust" OR "Zero Trust Security" OR "Zero-trust Security" (Topic) and Computer Science or Engineering (Research Areas) and Article or Proceeding Paper or Review Article (Document Types) and English (Languages) and 2025 or 2024 or 2023 or 2022 or 2021 or 2020 or 2019 or 2018 or 2017 or 2016 (Publication Years)

Similarly, in IEEE Xplore, we used:

("Document Title": "Zero Trust Architecture" OR "Document Title": "Zero-trust Architecture" OR "Document Title": "Zero-trust" OR "Document Title": "Zero Trust" OR "Document Title": "Zero Trust Security" OR "Document Title": "Zero-trust Security") OR ("Abstract": "Zero Trust Architecture" OR "Abstract": "Zero-trust Architecture" OR "Abstract": "Zero-trust" OR "Abstract": "Zero Trust" OR "Abstract": "Zero Trust Security" OR "Abstract": "Zero-trust Security") OR ("Index Terms": "Zero Trust Architecture" OR "Index Terms": "Zero-trust Architecture" OR "Index Terms": "Zero-trust" OR "Index

Table 1: **Inclusion Criteria**

Sn	Attribute	Description
1	Relevance to Research Questions	Articles must directly address one or more of the research questions.
2	Publication Type	Only peer-reviewed journal articles, conference proceedings, and reviews are included.
3	Publication Stage	Articles must have reached the "Final" publication stage to ensure completeness and credibility.
4	Language	Articles must be written in English to ensure consistency and comprehensibility.
5	Keywords and Content	Articles must explicitly mention or discuss "Zero Trust Architecture" or closely related terms such as "Zero Trust Security."
6	Time Frame	Publications must be from the years 2016 and 2025 to ensure the use of the most recent studies.

Table 2: **Exclusion Criteria**

Sn	Attribute	Description
1	Irrelevance	Articles that do not address the defined research questions or objectives, even if they mention Zero Trust Architecture.
2	Non-Peer-Reviewed Publications	Exclude technical reports, preprints, or non-peer-reviewed papers.
3	Non-English Articles	Articles written in languages other than English.
4	Publication Type	Exclude books, book chapters, editorials, news articles, and non-scholarly documents.

Terms": "Zero Trust" OR "Index Terms": "Zero Trust Security" OR "Index Terms": "Zero-trust Security") Filters Applied: Conferences Journals 2016 - 2025

The search progresses through several stages. Initially, we query the databases using the search terms only without applying any constraints. This yields 1,122, 539 and 642 articles from Scopus, WoS, and IEEE Xplore databases, respectively. We limit the search results by applying some filters, such as focusing on publications that reached final, written in English, from Computer Science and Engineering research area, while including only conference, article, and review papers. As a result, the number of retrieved articles is reduced to 839 articles in Scopus database, 485 articles in WoS database, and 458 in IEEE Xplore database. Finally, we restricted the search within a span of 10 years period from 2016 to 2025. This slightly reduces the results to 836, 482, and 456 articles from the Scopus, WoS, and IEEE Xplore databases, respectively. We believe the search strategy including the search keywords employed should capture all relevant publications to help us successfully answer our research questions.

4.4 SCREENING AND SELECTION PROCESS

After finishing the first search, it was clear that not every document that was found is directly related to the study's goals. As a result, a careful assessment of their applicability was necessary. This section outlined the four-stage screening and selection method used to choose the articles used in the study, as illustrated in Figure 4. The first step identifies and retrieves the research articles

based on the search terms described. We downloaded a CSV file that contains all the papers from our query response in the three databases. The file is a metadata of the retrieved articles that contains information about the papers including article titles, author full names, Keywords, abstract, DOIs, affiliations, document type, publication year, source title, and other pertinent information. The three downloaded CSV files were fed into a Python script that eliminated duplicate articles from the databases. We evaluated the articles for appropriateness and relevance in the second step of the process by looking at their titles and abstract in light of our objectives and research questions.

In the third step, we carry out a full-text screening to advance through the procedure. The inclusion and exclusion criteria listed in Tables 1 and 2 as well as the research questions were used to evaluate the papers. A publication must meet all inclusion requirements in order to be added to the review paper pool. Likewise, if a publication satisfies any of the exclusion criteria, it is not included in the pool. After eliminating the papers that either satisfied the exclusion criteria or did not meet the inclusion criteria, the final step of the process involved adding the resulting articles to the research paper pool.

4.5 STUDY SELECTION

The four step process of the PRISMA framework which are Identification, Screening, Eligibility, and Inclusion were used to select the articles for this study. Initially, after identifying and retrieving the research articles which resulted in a total of 1,774 research papers, 831 duplicate entries were eliminated using a Python script, 40 invalid entries usually from Scopus database, and 5 not available articles, resulting in 943 remaining articles. Following this, the articles were screened in the second stage based on their titles and abstracts. At this stage, 522 articles were removed by title, and 278 articles by abstract. 98 articles made it to the eligibility stage where we used full-text reading to assess them based on the inclusion and exclusion criteria and the research questions. Seven unrelated research articles were removed, leaving a total of 91 articles which were included in the SLR in the last stage of the process. These articles represent the primary source of information for the SLR.

4.6 Study Characteristics

This section describes some common characteristics of the studies included in this SLR. As depicted in Figure 5, 50% of the articles were journal publications, reflecting a strong preference for journals as the primary medium for publishing research on ZTA. , conference papers accounted for 42%, while review/survey papers represented 8%, indicating limited research on comprehensive reviews or surveys in the ZTA domain.

In addition, Table 3 presents the distribution of the articles based on source titles, however, only sources with more than one publication are shown. Among these, *IEEE Access* made the largest contribution with 12 publications, representing approxi-

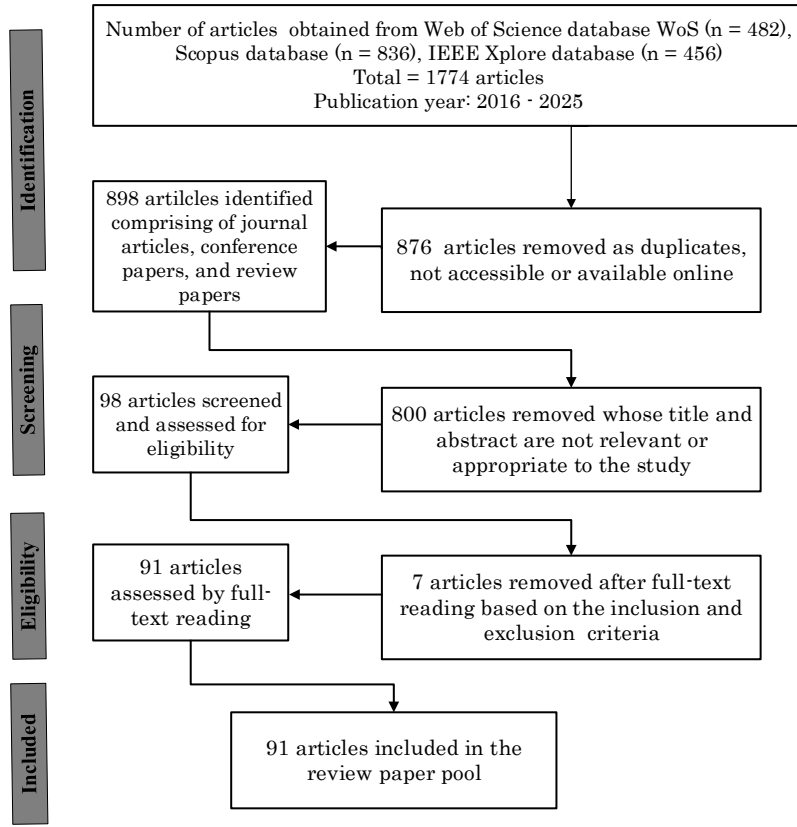


Figure 4: PRISMA Flowchart used in this study

Table 3: Distribution of the articles based on source title

Source Title	Reference	Number of Articles
IEEE Access	[14], [60], [61], [6], [1], [62], [63], [64], [26], [65], [66], [67]	12
IEEE Global Communications Conference, GLOBECOM	[68], [69], [70], [71], [72]	5
Electronics (Switzerland)	[73], [74], [75], [76]	4
Ad Hoc Networks	[4], [77], [78]	3
IEEE Wireless Communications	[79], [80]	2
Computer Networks	[81], [82]	2
Applied Sciences (Switzerland)	[83], [84]	2
Proceedings of IEEE/ACS International Conference on Computer Systems and Applications, AICCSA	[27], [85]	2
Wireless Communications and Mobile Computing	[86], [18]	2

mately 13% of the reviewed studies. The *IEEE Global Communications Conference, GLOBECOM* has five conference papers, followed by *Electronics (Switzerland)* and *Ad Hoc Networks* journals having four and three articles, respectively. Other sources,

including *IEEE Wireless Communications*, *Computer Networks*, *Applied Sciences (Switzerland)*, *Proceedings of IEEE/ACS International Conference on Computer Systems and Applications, AICCSA*, and the *Wireless Communications and Mobile Com-*

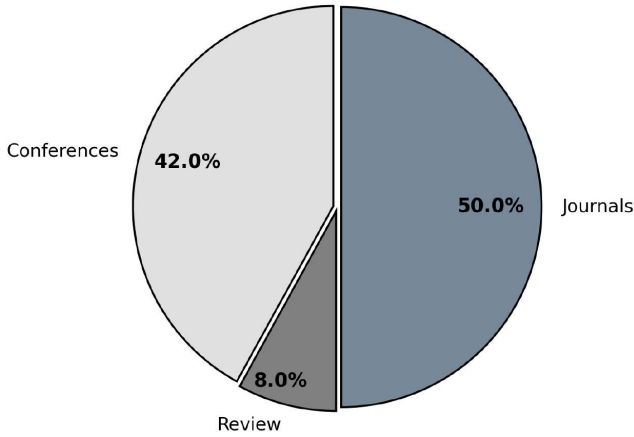


Figure 5: Distribution of the articles based on publication type

puting, each contributed two articles, collectively accounting for 11% of the total. The remaining 62% of the articles were distributed across a diverse set of journals and conferences, reflecting the broad application and multidisciplinary nature of Zero Trust Architecture.

5 Taxonomy

In this section we describe the proposed ZTA taxonomy developed in this SLR. Taxonomy is a systematic organization of components, concepts, or entities in a field or domain based on their shared characteristics and relationships. Thus, the ZTA taxonomy depicted in Figure 6 presents a structured way to explore ZTA by facilitating the identification of its various application domains, requirements, and enabling technologies. This classification not only simplifies the implementation process but also supports researchers and practitioners in analyzing ZTA across varying contexts.

5.1 Application Domains

Based on the existing literatures studied, it is evident that Zero Trust Architecture is being implemented across various sectors, each with unique security needs. In healthcare, the ZTA enables organizations to safeguard their patients' sensitive medical records and health data with robust access controls, secure data exchange protocols, and real-time monitoring of electronic health records (EHRs). ZTA in the finance sector helps to combat cyber threats like financial fraud through encrypted communications, strong identity verification, and network micro-segmentation. For the Internet of Things (IoT) including Industrial IoT, ZTA addresses the inherent vulnerabilities of interconnected devices by establishing secure identities, managing access permissions, and enforcing trust evaluations. In cloud computing, ZTA supports multi-tenant environments with dynamic access policies, real-time authentication, and precise micro-segmentation, safeguarding data and

applications in shared infrastructures. Likewise, as the 5G and 6G networks continue to evolve, ZTA offers greater security with continuous monitoring, dynamic trust evaluation, and distributed security policies designed for scenarios with high-speed, low-latency connectivity. The following subsections give a detailed discussion on these and other applications which indicate the applicability and increasing prevalence of ZTA security strategies in enhancing security defenses across diverse application domains.

5.1.1 Healthcare and Medical

The healthcare sector is a critical domain that relies heavily on computing devices especially with the proliferation of Internet of Health Things (IoHT) and handles diverse types of people data [87]. This data includes the health records or status of patients, their personal information, as well as financial details such as credit card information. The extensive devices used in the healthcare system expand its attack surface, while hosting large volume of sensitive data make it a high target for cyber attacks that can be executed if the system is compromised which can include identity theft, damage to reputation, and financial fraud [88]. Example of cyberattacks in the healthcare domain is the Conti ransomware attack on Ireland's Health Service Executive (HSE) which occurred in May 2021 [89]. The attack encrypted around 80% of the HSE's data, disrupted operations nationwide, made essential services such as radiotherapy to be suspended, and the hospitals had to revert to paper-based systems. Moreover, attacks on identity theft and reputational damage in the healthcare systems include the NationsBenefits data breach in 2023 where sensitive personal data of over 3 million individuals including social security numbers were exposed [90], and the Medibank breach in Australia in October 2022 which resulted in a major public relations crisis [91]. These incidents highlights the critical need for robust cybersecurity measures like the ZTA in the healthcare sector to protect sensitive data and maintain trust.

• Healthcare Systems

Healthcare systems are composed of diverse collection of devices including IoT devices, legacy systems, and unmanageable devices [84] whose interactions and inter-connectivity exceed beyond the local healthcare facility to the internet, for example, the cloud, exchanging private and control messages. While this presents so many advantages, it has equally expanded the attack surface [92]. This highlighted the need for a robust security to secure the healthcare systems. From the reviewed articles, Erik et al [23] conducted a research to assess the feasibility and practicality of adopting ZTA in healthcare institutions by comparing its perceived potentials against PBSM in terms of control, security, cost-effectiveness, risk, supportability, and operational aspects. They used a healthcare company based in the United States with a capacity of about 50K individuals as a case study. While qualitative analysis favor ZTA over PBSM due its increased security because of the architecture's micro segmentation requirement, but there are some associated overheads in terms of cost and support